Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 9

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Title

Implement Functional and Class Components in React

Aim

To design and develop a React application that demonstrates the implementation and integration of multiple functional components and a class component. The application contains separate reusable components such as Home, Student, Course, About, and Contact, along with a Faculty class component, and combines them through App.jsx.

OBJECTIVE

The objectives of this task are:

1. To understand the concept of components in React.
2. To create multiple functional components using JavaScript functions.
3. To create a class component by extending React.Component.
4. To organize React components into separate reusable files.
5. To import and render multiple components inside App.jsx.
6. To understand the use of JSX for creating user interfaces.
7. To demonstrate the difference between functional and class components.
8. To apply CSS styling to the components.
9. To understand component reusability and modular application structure.
10. To execute and verify the React application using a browser.

Theory

React is a JavaScript library used for building interactive and reusable user interfaces. A React application is divided into independent components, where each component represents a specific part of the user interface. Components make an application easier to develop, maintain, reuse, and understand.

A functional component is a JavaScript function that returns JSX. Functional components are the

modern and commonly preferred approach in React. They can use React Hooks such as useState and useEffect when state or lifecycle-related functionality is required.

A class component is created using a JavaScript class that extends React.Component. It normally uses a render() method to return JSX and can manage state through this.state and this.setState(). Class components were widely used in older React applications and remain useful when maintaining existing React projects.

In this experiment, separate functional components are created for Home, Student, Course, About, and Contact. A Faculty component is implemented as a class component. All components are imported and rendered from App.jsx, demonstrating modularity, component reuse, and the difference between the two component styles.

Concepts Used

Concept	Purpose
React Component	Represents an independent and reusable part of the user interface.
Functional Component	Creates a React component using a JavaScript function that returns JSX.
Class Component	Creates a React component using a class that extends React.Component.
JSX	Describes the structure and content of the user interface.
Props	Passes data from a parent component to a child component.
import	Makes components available for use in another file.
export default	Exports a component so it can be imported elsewhere.
render()	Returns JSX from a class component.
ReactDOM	Renders the React application into the HTML root element.
CSS	Provides styling and layout for the application.

ALGORITHM / PROCEDURE

1. Create a new React project using Vite.
2. Create a components folder inside the src directory.
3. Create Home.jsx, Student.jsx, Course.jsx, About.jsx, and Contact.jsx as functional components.
4. Create Faculty.jsx as a class component by extending React.Component.
5. Write JSX inside each component to display the required information.

6. Export each component using export default.

7. Import all components into App.jsx.

8. Render the functional and class components inside App.jsx.

9. Import App.jsx and index.css into main.jsx.

10. Apply CSS styling to the application using index.css.

11. Run the project using npm run dev.

12. Open the application in a web browser.

13. Verify that all functional components and the class component are displayed correctly.

14. Perform test cases to check component rendering and application output.

Code

App.jsx

```
import React from "react";

import Home from "./components/Home";
import Student from
"./components/Student"; import Course from
"./components/Course"; import About from
"./components/About"; import Contact from
"./components/Contact"; import Faculty
from "./components/Faculty";

function App() {
return (
<div className="app">
<h1>Student Management System</h1>

<Home />
<Student />
<Course />
<About />
<Contact />
<Faculty />
</div>
);
}

export default App;
```

Home.jsx

```
import React from "react";

function Home() {
return (
<div className="component">
<h2>Home</h2>
<p>Welcome to the Student Management System.</p>
<p>
This application demonstrates multiple functional
components in React.
</p>
</div>
);
}

export default Home;
```

Student.jsx

```
import React from "react";

function Student()
{ return (
<div className="component">
<h2>Student Details</h2>
<p><strong>Name:</strong> M. Niharika</p>
<p><strong>Roll Number:</strong> A24126552097</p>
<p><strong>Branch:</strong> CSE - AI & ML</p>
<p><strong>Year:</strong> 2nd Year</p>
</div>
);
}

export default Student;
```

Course.jsx

```
import React from "react";

function Course() {
return (
```

```
<div className="component">
<h2>Courses</h2>
<ul>
<li>Data Structures and Algorithms</li>
<li>Web Development</li>
<li>Database Management Systems</li>
<li>Artificial Intelligence</li>
<li>Machine Learning</li>
</ul>
</div>
);
}

export default Course;
```

About.jsx

```
import React from "react";

function About() {
return (
<div className="component">
<h2>About</h2>
<p>
This is a React application developed using
functional components.
</p>
<p>
Each section is implemented as a separate
reusable component.
</p>
</div>
);
}

export default About;
```

Contact.jsx

```
import React from "react";

function Contact()
{ return (
```

```
<div className="component">
<h2>Contact</h2>
<p><strong>Email:</strong> student@example.com</p>
<p><strong>Phone:</strong> +91 9876543210</p>
<p><strong>Location:</strong> Visakhapatnam</p>
</div>
);
}

export default Contact;
```

Faculty.jsx

```
import React, { Component } from "react";

class Faculty extends Component {
render() {
return (
<div className="component">
<h2>Faculty Details</h2>
<p><strong>Name:</strong> Dr. Anil Kumar</p>
<p><strong>Department:</strong> Computer Science</p>
<p><strong>Subject:</strong> Web Development</p>
</div>
);
}
}

export default Faculty;
```

main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";

import App from "./App";
import "./index.css";

ReactDOM.createRoot(document.getElementById("root")).render(
<React.StrictMode>
<App />
</React.StrictMode>
);
```

index.css

```
body {
margin: 0;
font-family: Arial, sans-
serif; background-color:
#f2f2f2;
}

.app {
text-align: center;
padding: 30px;
}

.app h1 {
color: #222;
margin-bottom: 30px;
}

.component {
background-color:
white; width: 70%;
margin:  20px
auto; padding:
25px; border-
radius: 10px;
box-shadow: 0 4px 10px rgba(0, 0, 0,
0.1); text-align: left;
}

.component h2 {
color: #007bff;
text-align:
center;
}

.component   p
{   font-size:
16px;    line-
height: 1.6;
}

.component  ul
{ line-height:
2;
```

```
}
```

CODE EXPLANATION

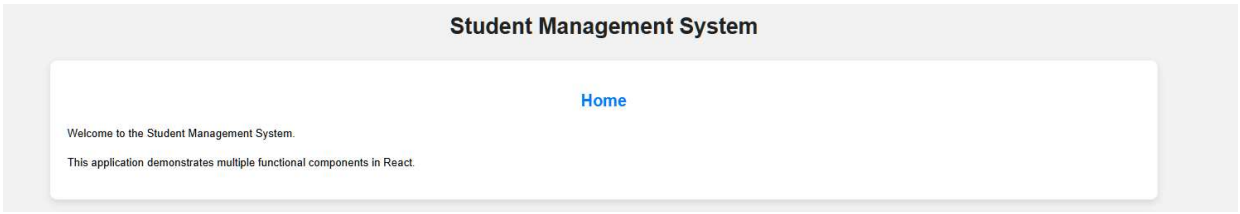
Code	Explanation
function Home()	Defines the Home functional component.
function Student()	Defines the Student functional component.
function Course()	Defines the Course functional component.
function About()	Defines the About functional component.
function Contact()	Defines the Contact functional component.
class Faculty extends Component	Defines the Faculty class component by extending React.Component.
render()	Returns the JSX displayed by the Faculty class component.
<Home />	Renders the Home functional component inside App.jsx.
<Student />	Renders the Student functional component.
<Course />	Renders the Course functional component.
<About />	Renders the About functional component.
<Contact />	Renders the Contact functional component.
<Faculty />	Renders the Faculty class component.
export default	Exports a component so that it can be imported into another file.
import	Imports a component or library into the current file.
ReactDOM.createRoot()	Creates a React root for rendering the application.
render()	Displays the App component in the root element.

TEST CASES AND EXECUTION DATA

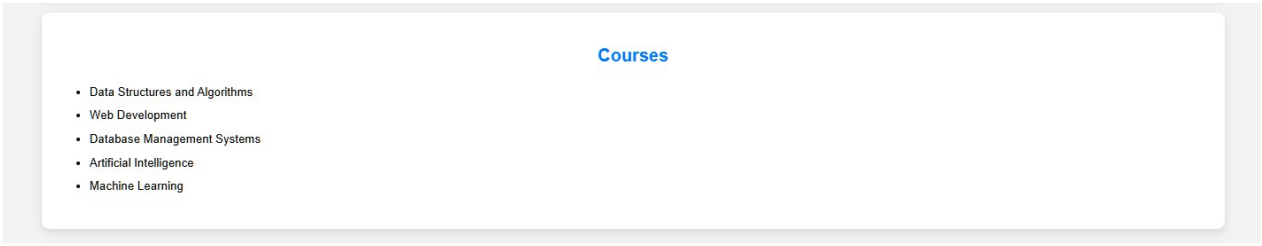
Test Case	Component Tested	Action / Input	Expected Result	Actual Result	Status
TC-01	Home	Run the React application.	Home component should be displayed.	Home displayed successfully.	Pass

Test Case	Component Tested	Action / Input	Expected Result	Actual Result	Status
TC-02	Student	Open the application and view Student section.	Student details should be displayed.	Student details displayed successfully.	Pass
TC-03	Course	View the Course section.	Course list should be displayed.	Course list displayed successfully.	Pass
TC-04	Faculty	View the Faculty section.	Faculty class component should be displayed.	Faculty displayed successfully.	Pass
TC-05	All Components	Run the complete application.	All components should render without errors.	All components rendered successfully.	Pass

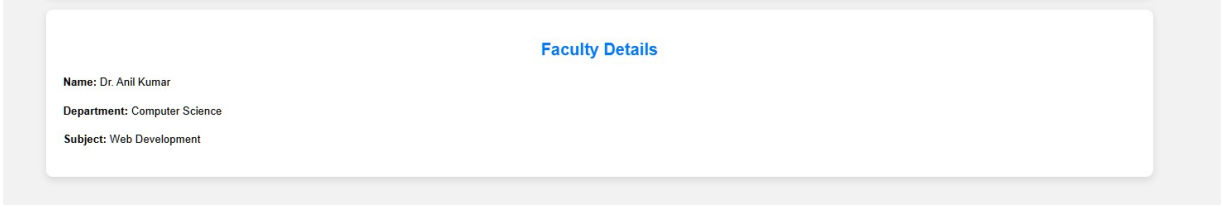
TC 01 & 02 Ouput :



TC 03 Ouput :

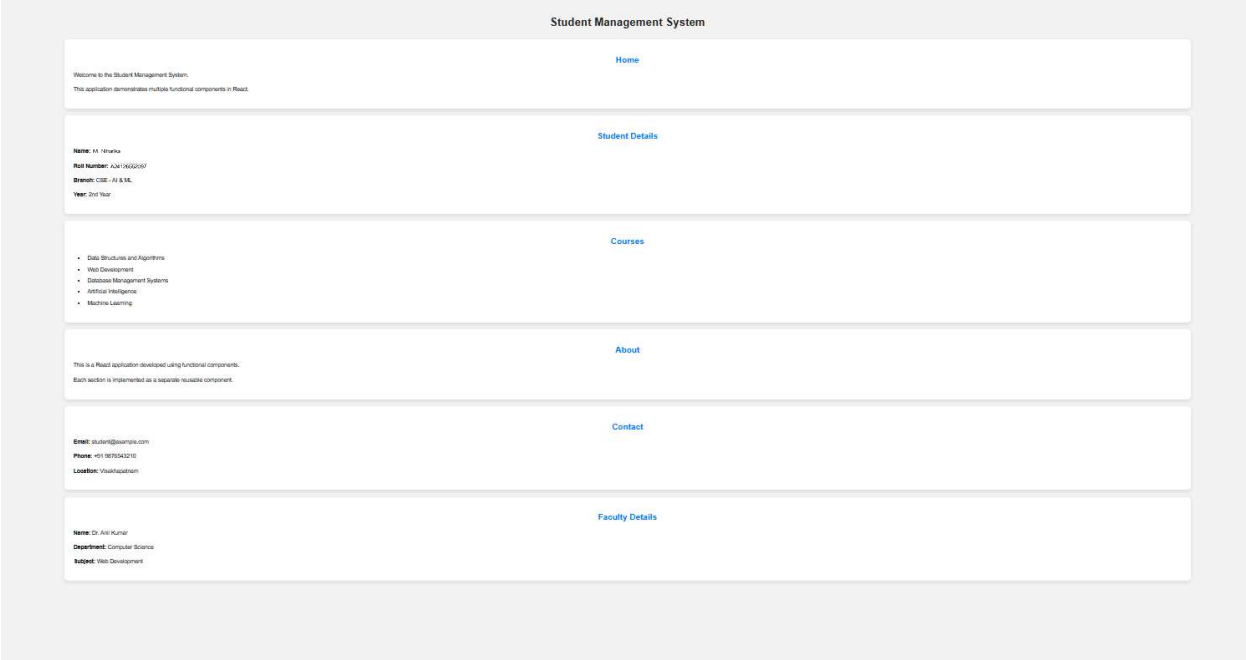


TC 04 Ouput :



Output

Output:



The browser displays the heading Student Management System followed by the Home, Student, Course, About, Contact, and Faculty sections. The first five sections are implemented as functional components and the Faculty section is implemented as a class component.

RESULT

Thus, a React application was successfully developed using multiple functional components and a class component. The Home, Student, Course, About, and Contact sections were implemented as functional components, while the Faculty section was implemented as a class component. All components were organized into separate files, imported into App.jsx, rendered successfully, and verified through the given test cases. The application demonstrates component reusability, modularity, JSX, importing and exporting components, and the difference between functional and class components.
